

Reading Statistics in Papers

LET Mathematics & Statistics

Example from Abu-Baker et al. (2021)

Descriptive statistics (e.g., means, standard deviations, frequencies, and percentages) were performed to describe the demographic characteristics of the participating students and the main study variables. For the belief scale, the two agreement categories (4 = agree, 5 = strongly agree) were collapsed to one category to indicate a positive belief. For the implementation scale, the three categories (2 = 4–5 times, 3 = 6–8, and 4 ≥ 8 times in the past 8 weeks) were collapsed to one category as (≥ 4 times) to indicate frequent implementation. Pearson's correlation test was used to determine the relationship between the total scores of the EBP belief and implementation scales. A chi-square test was used to examine the difference between trained and untrained students in terms of agreement toward each EBP belief (disagreement vs. agreement) and in terms of frequency of each EBP implementation (less than 4 times vs. 4 times or more in the past 8 weeks). Finally, an independent samples t-test was used to examine the difference between trained and untrained students in terms of the total mean scores of EBP beliefs.

1. The extract above describes how the researchers treated their data. Assuming the authors used the correct tests for their variables, identify the level of measurement (Nominal, Ordinal, or Interval/Ratio) for each of the following:
 - (a) Student agreement toward EBP beliefs (categorized as disagreement vs. agreement)
 - (b) The total scores of the EBP belief and implementation scales (analyzed using Pearson's correlation)
 - (c) The training status of the students (trained vs. untrained) as used in an independent samples t-test to split the groups

Solution

- a. The researchers 'collapsed' the categories into two named groups: disagreement vs. agreement. This makes the data Nominal.
- b. Total scores analyzed using Pearson's correlation require Interval/Ratio data.
- c. The grouping variable (trained vs. untrained) is Nominal.

Table 1: Distribution of the sample by demographic variables (n = 241)

Variables	n (%)
Gender	
Male	64 (26.6%)
Female	177 (73.4%)
Educational level	
Third-year	36 (14.9%)
Fourth-year	205 (85.1%)
University	
University A	123 (51%)
University B	118 (49%)
EBP training	
Yes	163 (67.6%)
No	78 (32.4%)
Total	241 (100%)

2. Referring to Table 1:

- (a) How many subjects in total were part of this study?
- (b) What percentage of the participants had received EBP training?
- (c) Approximately how many female participants were there for every male participant in this study
 - ☐ 1.5 females for every male
 - ☐ 1 male for every 3 males
 - ☒ **3 females for every male**

Solution

- The total sample size (n) is 241.
- According to the table, 163 students (67.6%) had received training.
- Divide the number of females (177) by the number of males (64). $177/64 \approx 3$.

Table 2: Responses to evidence-based practice belief scale by trained and untrained students (n = 241)

Your agreement to each statement:	All students	Trained	Untrained	χ^2	p
1. "I believe that EBP results in the best clinical care for patients".	177 (73.4%)	127 (77.9%)	50 (64.1%)	5.160	.023
2. "I am sure that evidence-based guidelines can improve clinical care".	174 (72.2%)	121 (74.2%)	53 (67.9%)	1.038	.308
3. "I am sure that implementing EBP will improve the care that I deliver to my patients".	167 (69.3%)	115 (70.6%)	52 (66.7%)	.374	.541
4. "I believe that critically appraising evidence is an important step in the EBP process".	153 (63.5%)	107 (65.6%)	46 (59.0%)	1.012	.314
5. "I believe the care that I deliver is evidence-based".	148 (61.4%)	102 (62.6%)	46 (59.0%)	.289	.591
6. "I am sure about how to measure the outcomes of clinical care".	152 (63.1%)	109 (66.9%)	43 (55.1%)	3.123	.077
7. "I am sure that I can implement EBP".	131 (54.4%)	99 (60.7%)	32 (41.0%)	8.261	.004
8. "I am clear about the steps of EBP".	127 (52.7%)	95 (58.3%)	32 (41.0%)	6.302	.021
9. "I believe that I can search for the best evidence to answer clinical questions in a time-efficient way".	144 (59.8%)	101 (62.0%)	43 (55.1%)	1.025	.311
10. "I believe that I can overcome barriers in implementing EBP".	124 (51.5%)	87 (53.4%)	37 (47.4%)	.745	.388
11. "I believe that EBP takes too much time".	125 (51.9%)	80 (49.1%)	45 (57.7%)	1.567	.211
12. "I am sure that I can access the best resources in order to implement EBP".	121 (50.5%)	83 (50.9%)	38 (48.7%)	.102	.749
13. "I believe EBP is difficult".	142 (58.9%)	90 (55.2%)	52 (66.7%)	2.858	.091
14. "I know how to implement EBP sufficiently enough to make practice changes".	128 (53.5%)	93 (57.1%)	35 (44.9%)	3.144	.076
15. "I am confident about my ability to implement EBP where I work".	129 (53.5%)	96 (58.9%)	33 (42.3%)	5.836	.016
16. "I am sure that I can implement EBP in a time-efficient way".	138 (57.3%)	99 (60.7%)	39 (50.0%)	2.485	.115

3. Referring to Table 2:

- (a) Identify the number of the statement shown above (1–16) that had the highest level of agreement when considering all students.
- (b) What is the percentage points difference between the trained and untrained students for Statement 1?
- (c) In almost every row of Table 2, the Trained group has higher agreement. However, for Statements 11 ('takes too much time') and 13 ('is difficult'), the Untrained group has higher agreement. How should a researcher interpret this?
- ☒ **It is a positive finding; the training successfully reduced the students' perception of these factors as barriers to EBP.**
- ☐ It is a negative finding; the training was unsuccessful because it failed to increase the level of agreement in the intervention group.
- ☐ It is an inconsistent finding; the reversal of the trend suggests that these two statements are less reliable than the other items.

Solution

- a. Statement 1 has 73.4% agreement across all students.
- b. $77.9\% - 64.1\% = 13.8$ percentage points.
- c. Statements 11 and 13 are negatively worded. Lower agreement in the trained group indicates a successful outcome (viewing EBP as less difficult/time-consuming).

Table 3: Scores of beliefs and implementations

Variable	Trained Mean (SD)	Untrained Mean (SD)	t	df	p	Mean Diff
Total score: beliefs	55.55 (13.29)	51.74 (14.06)	2.04	239	.042	3.80
Total score: implementations	27.01 (12.61)	21.86 (11.14)	3.08	239	.002	5.15

4. Referring to Table 3:

- (a) Identify the statistical test indicated by the column labeled 't'. What is it being used for here?
- ☐ Chi-square; comparing frequencies of agreement.
- ☒ **Independent samples t-test; comparing the mean scores of the two groups.**

- Paired samples t-test; comparing the mean scores between two time points.

Solution

The 't' column indicates an independent samples t-test, which compares the average (mean) scores of two separate groups.

- (b) Look at the row for 'Total score: beliefs.' Which of the following is the correct justification for whether the training made a significant difference?
- ⊗ **Yes; the p-value (.042) is less than 0.05.**
 - Yes; the t-value (2.04) is greater than 0.05.
 - No; the mean difference (3.80) is not large enough.

Solution

Statistical significance is determined by the p-value. Since $0.042 < 0.05$, the result is significant.

- (c) Compare the results for 'Beliefs' ($p = .042$) and 'Implementations' ($p = .002$). Which statement accurately summarizes the results?
- Only the implementation scores showed a significant difference.
 - Neither result is significant at the 5% level.
 - ⊗ **The training had a significant positive effect on both student beliefs and implementation frequency.**

Solution

Both p-values are less than 0.05, meaning both results are statistically significant.